MEGA SUPERMART

Workforce Exploratory Data Analysis

Final EDA Report | Employee dataset | 2025 absence-hours analysis

Purpose

This report explores the structure, quality and key patterns in Mega Supermart's anonymised employee dataset. It is designed to support management discussion about workforce composition, employee tenure and absence patterns across departments and stores.

1,200	1,198	38.6	4.54	2.52%
Employee records	Unique employee IDs	Average age (years)	Average tenure (years)	Weighted absence rate

Headline: The dataset contains 1,200 records and 1,198 distinct employee IDs. The overall weighted absence rate is 2.52%. Two employee IDs appear more than once, so employee-level totals should be interpreted with this data-quality issue in mind.

Prepared from the supplied anonymised employee CSV. The report uses the available fields and does not infer causes of absence.

1. Executive summary

The dataset provides a useful snapshot of workforce composition across 11 stores in 11 cities and 4 provinces. Sales is the largest department by employee records. The overall weighted absence rate is approximately 2.52%, calculated as total absent hours divided by total scheduled hours rather than averaging individual percentage values.

Key observations: Sales is the largest department (445 records); Customer Service has the highest department-level weighted absence rate (3.38%); and ST10 (Regina) has the highest store-level weighted absence rate (5.09%).

The average employee age is 38.6 years and average length of service is 4.54 years. There are 247 records with less than one year of service and 99 records for employees aged 55 or older.

These findings identify areas for further investigation; they do not establish why absence rates differ. Differences may reflect workforce mix, scheduling, local operating conditions, recording practices or other factors not captured in the dataset.

2. Dataset overview

Item	Result
Rows / employee records	1,200
Columns in source file	17
Distinct EmployeeID values	1,198
Stores / cities / provinces	11 / 11 / 4
Scheduled hours total	2,224,144
Absent hours total	55,984
Weighted absence rate	2.52%

The source contains employee attributes, organisational structure, hire and birth dates, age, length of service, scheduled hours, absent hours and a supplied absence-rate field.

3. Data quality assessment

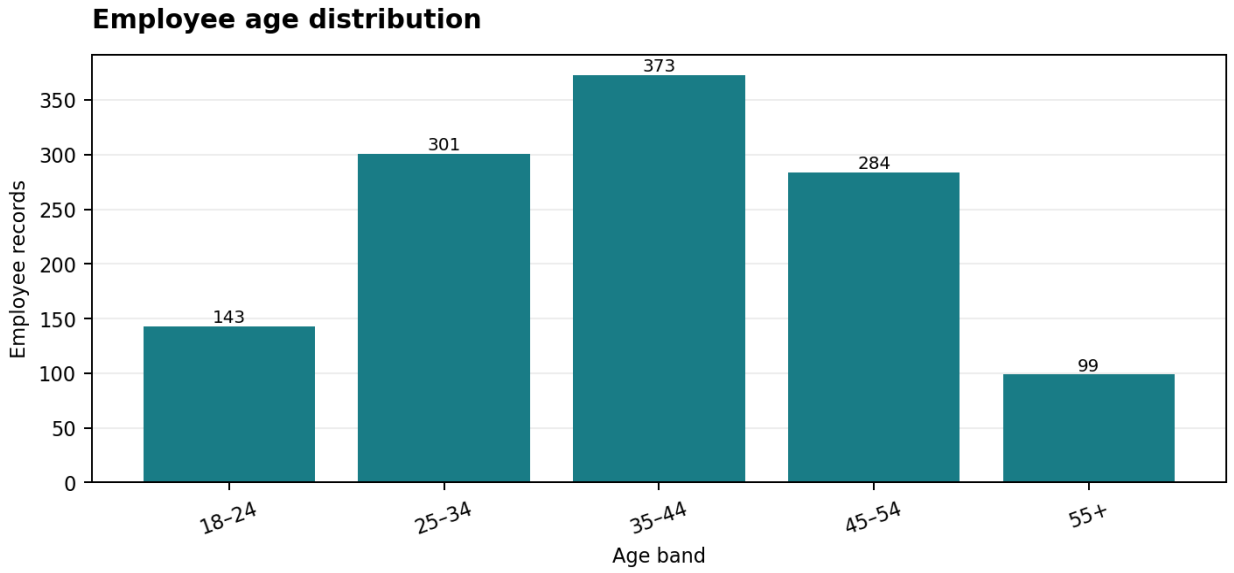
A quality review was completed before interpreting the workforce and absence patterns. The checks below are important because duplicated employee IDs and missing values can affect counts and rate calculations.

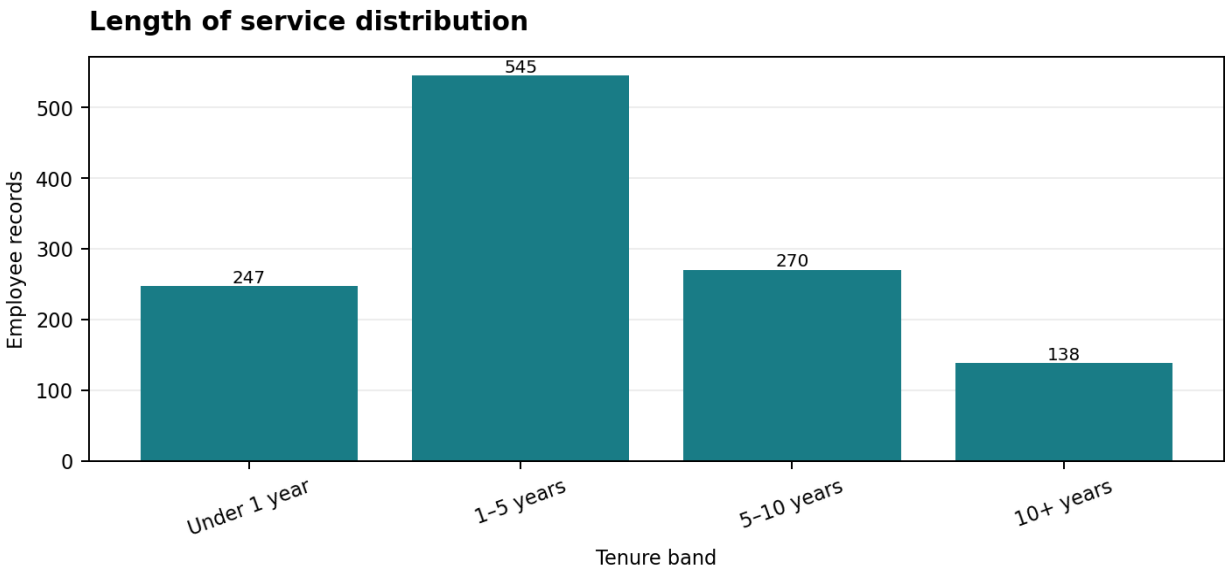
Check	Result	Interpretation
Missing Gender	3	Three records have no gender value.
Missing AbsentHours	2	Two records lack absent-hours data; totals exclude these missing values.
Missing supplied AbsenceRate	2	Two records lack the supplied percentage; weighted rate is recalculated from hours.
Repeated EmployeeID values	2 IDs / 4 rows	Four rows are associated with repeated IDs; investigate before employee-level reporting
Exact duplicate rows	0	No fully identical rows were detected.
Department naming variations	2 variants	“sales” and “Customer service” were standardised for grouping.

Recommended treatment: verify the four rows linked to repeated employee IDs against the source system. Do not automatically delete them until it is confirmed whether they are accidental duplicates or legitimate multiple records. Standardise category spelling and case, and decide how missing gender and absence values should be handled for each analysis.

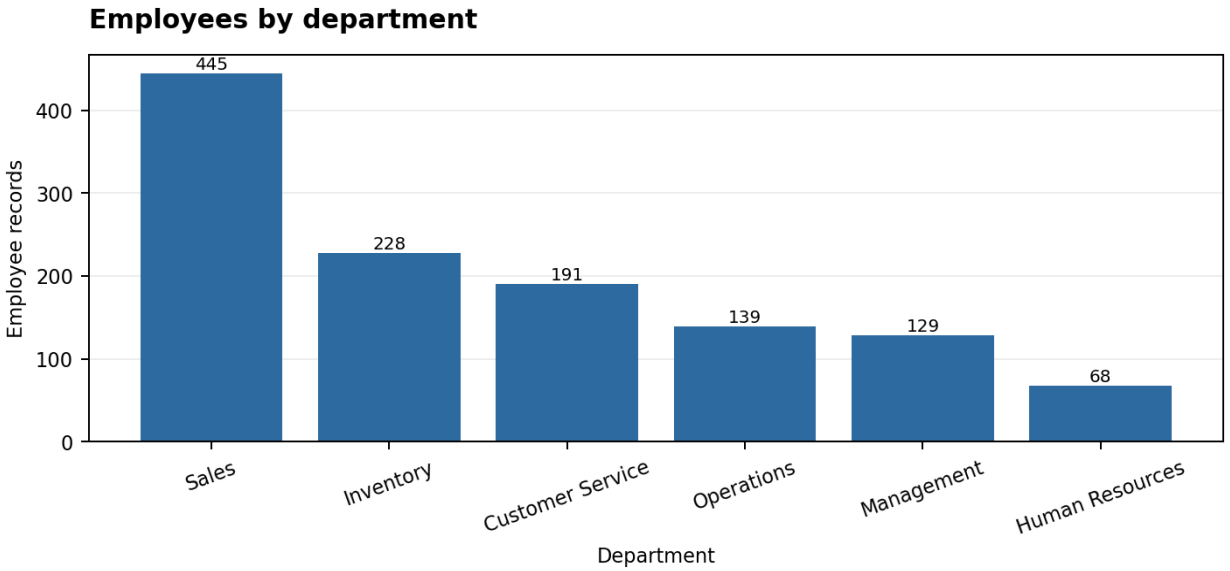
4. Workforce profile

The mean age is 38.6 years (median 38); the mean length of service is 4.54 years (median 3.06). These averages describe the records in the supplied file and may be affected by repeated employee IDs.

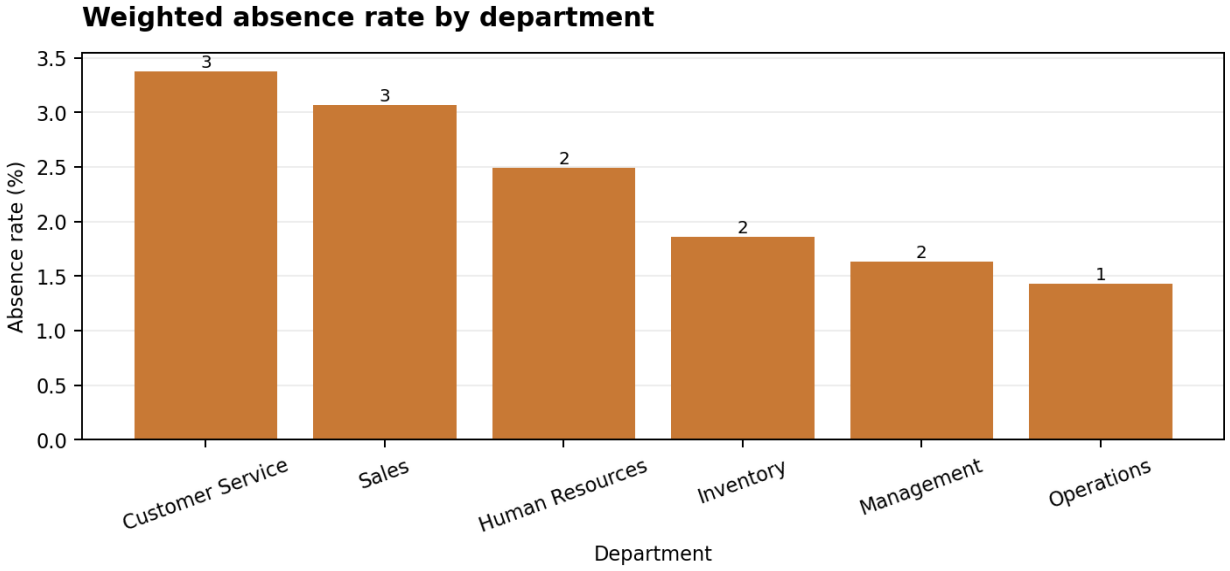



Length of service distribution
Employee records
0
100
200
300
400
500
247
545
270
138
Under 1 year
1–5 years
5–10 years
10+ years
Tenure band

5. Department analysis

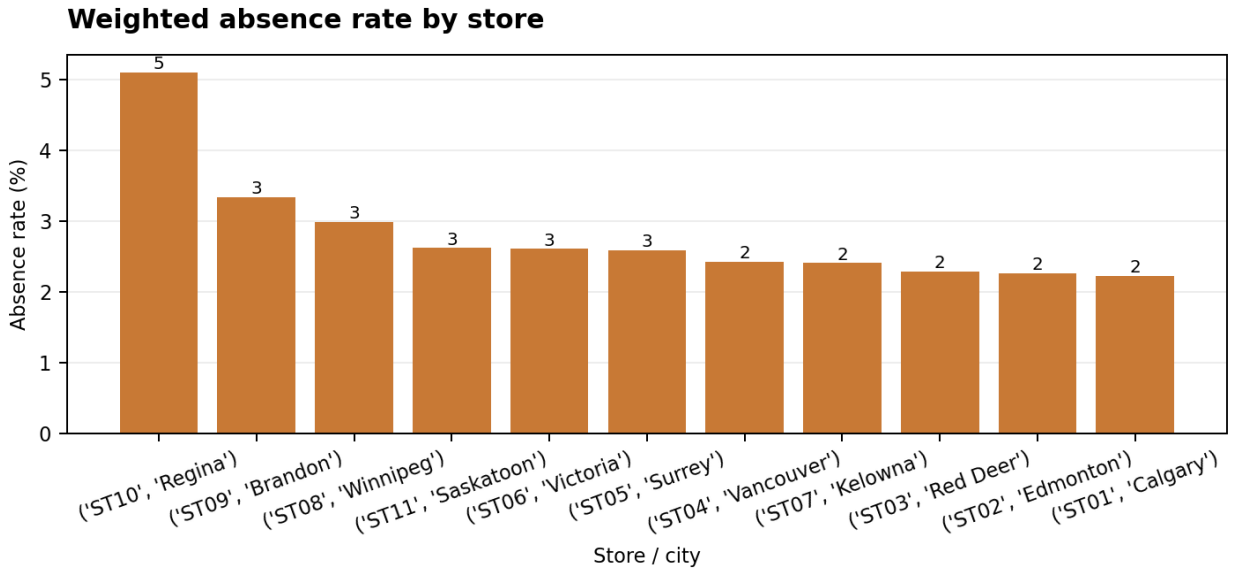


Sales has the largest recorded headcount (445). Customer Service and Inventory are also substantial departments. These headcount patterns can help management contextualise staffing and operational planning, but staffing adequacy cannot be assessed without workload, sales volume, shift coverage and productivity data.

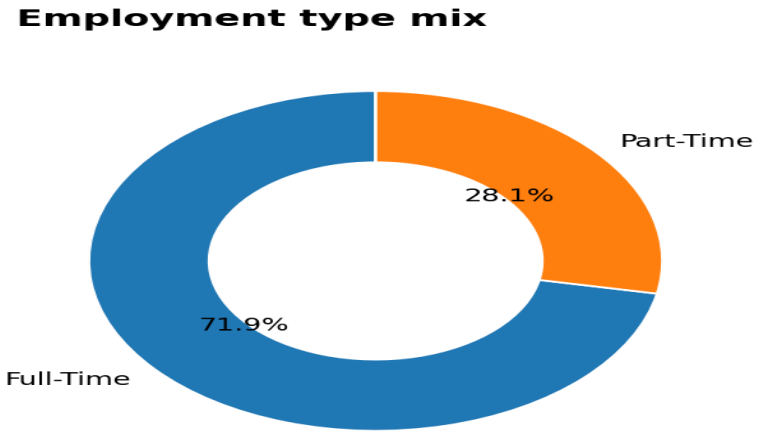


Customer Service and Sales have the highest weighted department absence rates in this dataset. Operations and Management have the lowest. A higher rate is a signal for further review, not proof of poor attendance or a management problem.

6. Store and employment analysis

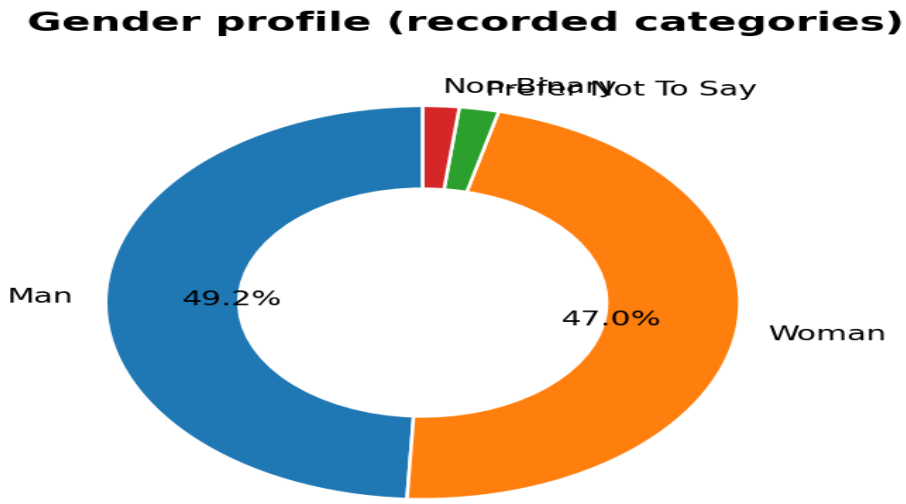


ST10 in Regina records the highest weighted absence rate (5.09%); the lowest is ST01 in Calgary (2.22%). Store-level differences should be checked alongside staffing levels, scheduled hours, local conditions and consistency of absence recording.



The dataset contains 863 full-time and 337 part-time records. The weighted absence rate is 2.22% for full-time records and 3.75% for part-time records. These rates should be interpreted in light of differences in scheduled hours and employee mix.

7. Gender profile



Gender values include Man, Woman, Non-Binary and Prefer Not To Say, with three missing entries. These categories are reported as supplied and should not be combined or reclassified. Use this breakdown only where it serves a clear, appropriate workforce-planning purpose.

8. Key findings and management implications

Finding	Management implication / question
Overall weighted absence rate: 2.52%	Use as a baseline and monitor consistently over time, ideally by month and shift.
Highest department rate: Customer Service (3.38%)	Review absence patterns, shift coverage, workload and local context with department leaders.
Highest store rate: ST10 – Regina (5.09%)	Validate the underlying records and compare with other stores over a consistent period.
247 records have under one year of service	Review onboarding, induction, early-tenure support and retention indicators.
99 records are aged 55 or older	Consider succession planning, skills transfer and workforce capability planning without making as

9. Recommendations

• **Resolve identity duplicates:** investigate repeated EmployeeID values and document the approved deduplication rule before publishing final headcount.

• **Standardise data entry:** use consistent department labels and controlled category values across stores.

• **Monitor absence monthly:** calculate absence as total absent hours divided by total scheduled hours, and show both numerator and denominator.

• **Investigate outliers carefully:** review unusually high absent-hour values, including the maximum recorded value of 720 hours, to confirm period, units and validity.

• **Support early-tenure employees:** examine onboarding, training and retention patterns for employees with less than one year of service.

• **Use context before action:** investigate high-rate departments and stores with operational leaders before making staffing or performance decisions.

• **Improve the dataset:** add month, shift, absence reason (where lawful and appropriate), FTE, scheduled days and workload measures to enable deeper analysis.

10. Methodology and limitations

Absence-rate calculation: weighted absence rate = sum of AbsentHours ÷ sum of ScheduledHours2025. This approach weights each record by scheduled hours and is preferable to taking a simple average of the supplied AbsenceRate percentages for an overall or group rate.

Data preparation: numeric fields were treated as numeric values; department labels were trimmed and normalised for grouping; missing values were not imputed. Gender categories were retained as supplied. Counts in this report refer to records unless explicitly described as distinct EmployeeID values.

Limitations: the data is a single supplied snapshot with no monthly trend, absence reasons, shift details, FTE information, workload or business performance measures. Repeated employee IDs remain unresolved. Therefore, the analysis is descriptive and cannot establish causation or support conclusions about individual employee performance.

Important note: The supplied AbsenceRate column is stored as text in percentage format. For Power BI, use a DAX measure based on summed AbsentHours divided by summed ScheduledHours2025, then format the measure as a percentage.

Appendix: Power BI measure

Use this measure in Power BI to reproduce the overall and filtered absence rate:

```
Overall Absence Rate =
DIVIDE(
SUM(employee_clean[AbsentHours]),
SUM(employee_clean[ScheduledHours2025]),
0
)
```

End of report